

MICROBIOLOGY

CRISPRi-based functional genomic screening identifies genes essential for CH₄-dependent growth in a methanotrophic bacterium

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Methanotrophic bacteria are the primary organisms that consume atmospheric methane (CH₄) and have potential to mitigate climate-active gases. However, a limited understanding of the genetic determinants of methanotrophy hinders the development of biotechnologies leveraging these unique microbes. Here, we developed and optimized a CRISPR interference (CRISPRi) system to enable functional genomic screening in methanotrophic bacteria. We built a genome-wide single guide RNA (sgRNA) library in the industrial methanotroph, *Methylococcus capsulatus*, consisting of ~45,000 unique sgRNAs mediating inducible, CRISPRi-dependent transcriptional repression. A selective screen during growth on CH₄ identified genes associated with essential methanotrophic bacterial processes and previously unidentified essential gene candidates, highlighting the utility of CRISPRi for functional genetic screening in methanotrophs. The CRISPRi screen also led to nitrate mineral salts medium optimization to improve methanotroph biomass productivity from CH₄. Collectively, our results show that the CRISPRi system and sgRNA library developed here can be used for facile gene-function analyses and genomic screening to identify genetic determinants of methanotrophy and isolation of improved methanotroph biocatalysts.

INTRODUCTION

Aerobic methanotrophic bacteria, methanotrophs, play an essential role in balancing atmospheric carbon by oxidizing methane (CH₄) to multicarbon molecules (methanotroph biomass) and carbon dioxide (CO₂). These ubiquitous microbes limit the escape of CH₄ to the atmosphere via biofiltration in terrestrial and aquatic systems (1). Methanotrophs can also be leveraged for the bioconversion of abundant CH₄-rich gas streams from oil and natural gas extraction sites, coal mines, agricultural waste, landfills, and wastewater treatment facilities and have the potential to capture rising CH₄ greenhouse gas and mitigate climate change (2). However, substantial advances in methanotroph biology and engineering for effective CH₄ capture and conversion are needed to realize cost-effective solutions using methanotrophs.

Methanotrophs belong to a broad group of methylotrophic microbes that use organic single carbon (C₁) molecules as sources of carbon and/or energy, which, in addition to CH₄, includes methanol (CH₃OH), formate (CH₂O), and/or methylamine. Unique to methanotrophs is the CH₄ monooxygenase (MMO) enzyme that oxidizes the strong C–H bond of CH₄ to generate CH₃OH (3). Methanotroph C₁ metabolic enzymes and pathways for CH₃OH assimilation to biomass and its dissimilation to CO₂ generally overlap with those in methylotrophs (4). These include the ribulose monophosphate and serine cycles used by gammaproteobacterial methanotrophs and alphaproteobacterial methanotrophs for C₁ assimilation, respectively (5–7). In addition, the methanotroph dissimilatory metabolic processes that couple C₁ oxidation to the electron transport chain for ATP (adenosine triphosphate) production are coincident with those in other methylotrophs (8, 9). These central CH₄ metabolic pathways have been biochemically validated using radiolabeled isotopes in a subset of

methanotrophs (5, 7, 10), and the availability of several complete methanotroph genomes and comparative genomics have provided critical insight into methanotroph CH₄ metabolism (11–18). However, many genes that may be required for methanotrophy, especially those outside of the core C₁ metabolic gene clusters, are difficult to identify due to methanotrophs' strict CH₄-dependent cultivation.

Early research using chemical and insertional transposon mutagenesis of the CH₃OH-utilizing methylotrophic bacterium *Methylobacterium extorquens* AM1 identified many genes required for methylotrophic growth, including those encoding the enzymes needed for methanol oxidation, formaldehyde oxidation, formate oxidation, and assimilation via the serine cycle (19–26). In addition, transposon mutagenesis led to the identification of genes required for transport and utilization of rare earth elements (27) and transcriptional regulators of the C₁ metabolic pathways (20, 28). Notably, the elucidation of genes required for methylotrophy was enabled by leveraging their facultative capacity to use alternative multicarbon sources (i.e., succinate) supplied during library generation (28). In contrast, the vast majority of methanotrophs are obligate CH₄-utilizers; thus, mutations in essential genes required for growth on CH₄ would be lost from the mutagenized population. There are only two reports of successful chemical mutagenesis and one report of transposon insertional mutagenesis in methanotrophs (29–31), underscoring barriers applying forward genetic techniques to CH₄-oxidizing bacteria and substantial knowledge gaps related thereto.

Clustered regularly interspaced short palindromic repeats (CRISPR) interference (CRISPRi) is a genetic tool that has been leveraged for functional genetic screens in diverse bacteria (32–40). CRISPRi uses the nuclease-deficient “dead” CRISPR-associated (Cas) variant dCas9 and a single guide RNA (sgRNA) for the sequence-specific repression of gene expression (41). This technology can be used to target single genes or scaled for genome-wide screening via massively parallel oligonucleotide synthesis of pooled sgRNAs. Functional genomic screens using CRISPRi are coupled with next-generation sequencing to

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quantify the depletion or enrichment of sgRNAs (sgRNA differential fitness) during selective bacterial cultivation (42). Notably, transcription repression by CRISPRi can be tightly regulated using inducible promoters to drive dCas9 and/or sgRNA expression; thus, in contrast to transposon mutagenesis, CRISPRi can be applied to conditionally evaluate the effect of repressing essential gene expression on bacterial growth. Recent developments of methanotroph CRISPR genome editing tools indicate that Cas9 and sgRNAs are functional in gammaproteobacterial and alphaproteobacterial methanotrophs (32, 43–47), paving the way for CRISPRi and forward genetic approaches in methanotrophs that have eluded these bacteria for decades.

Here, we modified a previously developed methanotroph CRISPR-Cas9 genome editing system (45) to facilitate CRISPRi and demonstrate its utility for repressing gene transcription in the phylogenetically diverse methanotrophs *Methylococcus capsulatus* Bath and *Methylosinus trichosporium* OB3b. We performed a genome-scale CRISPRi screen using 45,798 unique sgRNAs targeting all annotated genes in the *M. capsulatus* Bath genome to investigate gene-function relationships during CH₄ cultivation. We identified 233 genes whose transcription repression resulted in significantly decreased fitness and repression of 13 genes associated with a fitness advantage. sgRNAs with decreased fitness scores targeted genes encoding proteins involved in critical cellular processes like ribosome biosynthesis, translation, transcription, and central biosynthetic metabolism, underscoring the utility of CRISPRi for the identification of predicted and previously unidentified essential genes. The CRISPRi system and sgRNA library developed here can be used for facile gene-function analyses

and genomic screening to identify methanotroph genetic determinants under an array of screening conditions. Furthermore, these CRISPRi screening methodologies can also be applied to high-throughput engineering approaches for isolation of improved methanotroph biocatalysts.

RESULTS

Construction and validation of methanotroph CRISPRi tools

We previously developed a two-plasmid, broad-host-range CRISPR-Cas9 system that enabled *M. capsulatus* genome editing (45). This two-plasmid system consisted of a (i) pBBR-based plasmid with the *M. capsulatus* constitutive PQQ-dependent methanol dehydrogenase *P_{mxhF}* promoter driving sgRNA expression and a compatible (ii) IncP-based plasmid, pCAH01, with anhydrotetracycline (aTc)-inducible Cas9 expression (pCAH01^{SpR}::cas9) (45). To construct a single-plasmid CRISPRi system, we moved the *P_{mxhF}*-sgRNA module to pCAH01^{SpR}::cas9 and replaced the *cas9* gene with a *dcas9* gene encoding the nuclease-deficient dCas9 variant, creating pJHCi (Fig. 1A). The ability of the CRISPRi system to repress transcription in the gammaproteobacterial methanotroph *M. capsulatus* Bath and alphaproteobacterial methanotroph *M. trichosporium* OB3b was tested using strains expressing a monomeric red fluorescent protein (mRFP) reporter (Fig. 1B) (48). pJHCi with an sgRNA targeting the nontemplate strand of the mRFP open reading frame (sgRNA^{mRFP}) was transferred to the methanotroph reporter strains via conjugation. The CRISPRi system reduced *M. capsulatus* and *M. trichosporium* mRFP

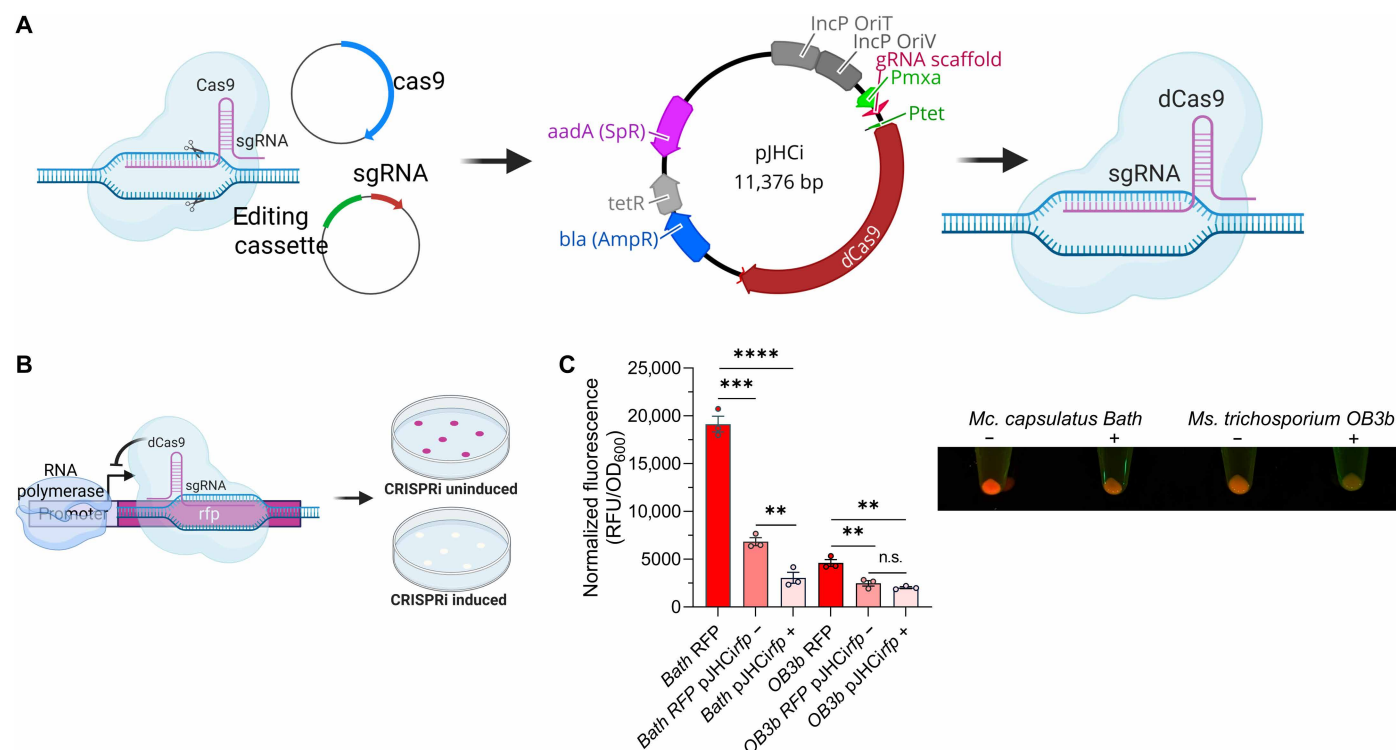


Fig. 1. Construction and validation of a methanotroph CRISPRi system. (A) Schematic overview of converting a two-plasmid CRISPR genome editing system into a single-plasmid CRISPRi system for targeted repression of gene expression. (B and C) The mRFP fluorescent reporter was used as a readout of CRISPRi functionality in gammaproteobacterial methanotroph *M. capsulatus* Bath and alphaproteobacterial methanotroph *M. trichosporium* OB3b with (+) or without (-) dCas9 induction by aTc. A representative image of cell pellets is shown. RFU, relative fluorescence units. Data in (C) represent the mean $\pm$ SEM from two independent experiments ($n = 3$ to 5). ** $P \leq 0.01$; *** $P \leq 0.001$; **** $P \leq 0.0001$; n.s., not significant. (A) and (B) were created in BioRender. Henard, C. (2026) <https://BioRender.com/chvt7zp>; <https://BioRender.com/7yozn5v>.

fluorescence by 6.3- and 2.3-fold, respectively, after 72 hours of bacterial cultivation in liquid nitrate mineral salts (NMS) with CH₄ as the carbon source (Fig. 1C). The decreased efficiency of the CRISPRi system in *M. trichosporium* may be related to differences in sgRNA and/or dCas9 abundance as promoters derived from *Gammaproteobacteria*, like those driving transcription of the sgRNA (*M. capsulatus* P_{mx}) and dCas9 (P_{tet}), exhibit lower activity in *M. trichosporium* compared to *M. capsulatus* (48).

The pJHCi CRISPRi system limited the production of fluorescent protein in the absence of dCas9 induction in both methanotrophs (Fig. 1C), suggesting that low dCas9 expression from leaky P_{tet} promoter gene transcription in the context of high, constitutive sgRNA expression is sufficient for transcription repression. The leaky functionality of the CRISPRi system is not desirable for functional genomic applications where sgRNA expression could cause fitness defects during strain or library generation. Thus, to develop a system with more tightly regulated CRISPRi repression, we designed a configuration wherein both dCas9 and the sgRNA are inducible (Fig. 2A). mRFP fluorescence in this strain was similar to a control strain with the mRFP reporter plasmid alone (Fig. 2B), supporting that the CRISPRi system did not inhibit mRFP transcription in the absence of aTc induction. In contrast, simultaneous induction of dCas9 and sgRNA^{mRFP} led to a 3.3-fold decrease in mRFP fluorescence (Fig. 2B). Fluorescence microscopy was used to assess CRISPRi efficiency in individual cells. As shown in Fig. 2C, CRISPRi repressed mRFP fluorescence in most of the bacterial population, but some

cells exhibited high mRFP fluorescence, indicating a small percentage of the population may “escape” CRISPRi repression.

CRISPRi-mediated transcription repression of *M. capsulatus* RuBisCO causes growth inhibition

We previously showed that the ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO) enzyme encoded by *M. capsulatus* is essential for its growth on CH₄ (49). As such, we targeted the *cbbL* gene encoding the RuBisCO large subunit to test the ability of the CRISPRi system to repress transcription from the *M. capsulatus* chromosome. Transconjugants could only be obtained in the absence of sgRNA^{cbbL} and dCas9 induction (Fig. 3A), supporting the essentiality of RuBisCO for *M. capsulatus* growth (49). Transconjugants isolated in the absence of induction were subsequently cultivated in liquid NMS with or without the aTc inducer, and high-resolution growth analysis showed that CRISPRi-repression of *cbbL* transcription significantly inhibited bacterial growth (Fig. 3B). Notably, the induced culture began to expand after 5 days and exhibited typical logarithmic growth kinetics thereafter (Fig. 3B). The aTc inducer has a half-life of 17 hours in minimal medium at pH 7 and 37°C (50), so we hypothesized that the delayed growth observed in induced samples may be due to ineffective aTc concentrations after 5 days of cultivation at 37°C. However, we did not observe a growth defect when the induced, outgrown sgRNA^{cbbL} culture was diluted into fresh induction medium, indicating that the CRISPRi system is no longer effective in this expanded population. Known mutations

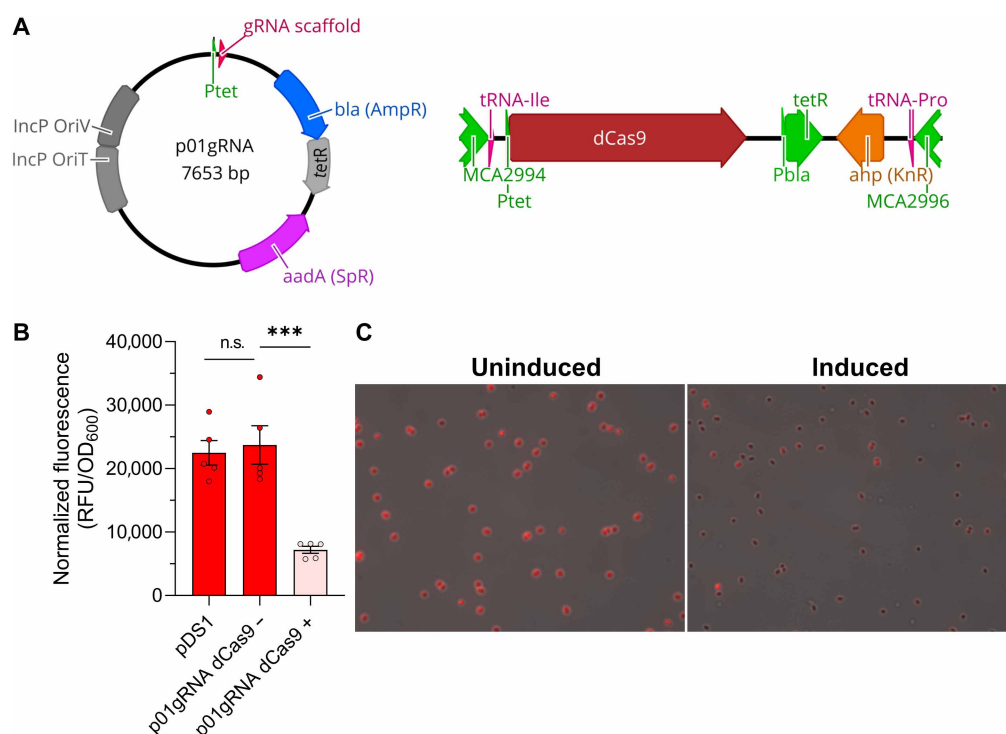


Fig. 2. A CRISPRi system with inducible dCas9 and sgRNA enables fine-tuned transcription repression in *M. capsulatus*. (A) The pJHCi single-plasmid system exhibited leaky functionality, so an *M. capsulatus* Bath strain was generated with an aTc-inducible dCas9 cassette integrated into the chromosome and plasmid with aTc-inducible sgRNA expression. (B) mRFP fluorescence with or without aTc induction compared to the pDS1 mRFP reporter plasmid control strain showing tightly regulated CRISPRi gene transcription repression. (C) Fluorescence microscopy of *M. capsulatus* with or without CRISPRi induction for mRFP repression. Representative images from biological replicates are shown. Data in (B) represent the mean $\pm$ SEM from two independent experiments ($n = 3$ to 5). n.s., not significant; *** $P \leq 0.001$. (A) was created in BioRender. Henard, C. (2026) <https://BioRender.com/eeb09r>.

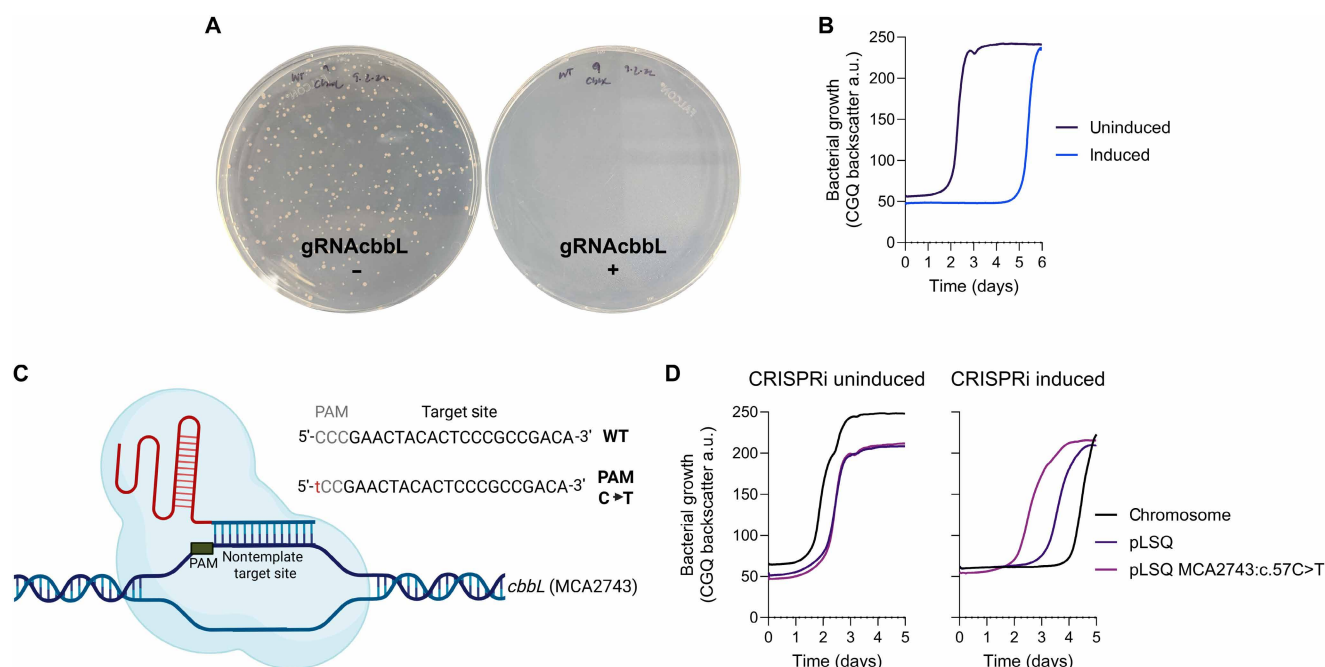


Fig. 3. CRISPRi-mediated transcription repression of the genes encoding the essential RuBisCO enzyme limit *M. capsulatus* growth. Growth of *M. capsulatus* with (+) or without (–) induction of dCas9 and an sgRNA targeting the *cbbL* gene encoding the large subunit of RuBisCO on solid NMS medium (A) or in liquid NMS medium in a serum vial over time (B) with 20% CH₄ in air atmosphere. (C) A complementing plasmid expressing the *cbbLSQ* operon (pLSQ) was constructed, and the PAM site in the *cbbL* target sequence (MCA2743) was mutated (c57C>T). WT, wild type. (D) Growth of strains carrying the pLSQ or pLSQ^{ΔPAM} compared to an isogenic background control strain with or without CRISPRi induction. a.u., arbitrary units. Data in (B) and (D) represent the mean of two independent high-resolution cultivation experiments ($n = 6$). (C) was created in BioRender. Henard, C. (2026) <https://BioRender.com/n4w9sga>.

in the dCas9 gene (51) or others in the sgRNA, *Ptet-tetO* promoter, or the *tetR* gene that would render the CRISPRi system nonfunctional were not identified in five “escapee” colonies via Sanger sequencing, so the underlying mechanism of resistance to CRISPRi repression in these strains remains elusive. Given the essentiality of RuBisCO for *M. capsulatus* growth, it is likely there is a strong selective pressure to overcome *cbbL* transcription repression in these cells.

To further test the specificity of the CRISPRi system, we constructed a plasmid containing the *cbbLSQ* operon under control of its native promoter region (pLSQ) and another plasmid with a mutated *cbbL* target protospacer adjacent motif (PAM) (MCA2743:c.57C>T, pLSQ^{ΔPAM}) expected to prevent dCas9 DNA binding (52, 53) (Fig. 3C). CRISPRi targeting of *cbbL* in the pLSQ^{ΔPAM} strain did not inhibit growth (Fig. 3D), supporting that expression of RuBisCO from this plasmid was not inhibited and able to complement chromosomal *cbbLSQ* transcription repression. In contrast, CRISPRi-mediated repression of *cbbL* transcription in the strain with pLSQ inhibited bacterial growth, but not to the extent of that observed in the strain lacking the plasmid (Fig. 3D). The pLSQ plasmid has a pBBR-based replicon, which is maintained in *M. capsulatus* at ~50 copies per cell (48). As such, the difference in growth inhibition observed between the pLSQ and control strain may be correlated to an increase in target sites in cells harboring pLSQ, thereby limiting CRISPRi transcription repression efficiency compared to targeting a single chromosomal site.

CRISPRi functional genomic screening identifies genes essential for CH₄-dependent growth

To evaluate the utility of our CRISPRi system for functional genomic screening, we designed and synthesized a pooled sgRNA library

consisting of 45,798 sgRNAs, ~15 sgRNAs per *M. capsulatus* protein-coding genes (3034 annotated in the NC_002977.6 RefSeq genome), and 400 nontargeting control sgRNAs (tables S1 and S2). Eighteen open reading frames had less than 15 sgRNAs in the library, and zero sgRNA target sequences were identified for three genes (MCA0050, MCA0307, and MCA1517; table S3). The workflow for cloning and screening the sgRNA library is shown in Fig. 4A. We determined the *M. capsulatus* library coverage by amplifying the sgRNAs from genomic DNA isolated from 6.8×10^7 pooled transconjugants. The sgRNA amplicons were sequenced using an Oxford Nanopore Technologies (ONT) MinION device; 680,425 reads with QC 12+ scores were trimmed and filtered using Geneious software followed by extraction of the 20–base pair (bp) sgRNA targeting sequences from the processed sequences using a custom script. In total, 680,304 extracted sgRNA spacers were aligned to the synthesized genome-wide CRISPRi sgRNA library reference with 45,554 of the 45,798 synthesized sgRNAs detected (99.5% sgRNA library coverage; table S4) with at least one sgRNA per predicted coding DNA sequence (except the three genes originally not included in the synthesized library; table S5).

The methanotroph genome-wide sgRNA library was cultivated in serum vials with an atmosphere of 20% CH₄ in air with and without CRISPRi induction to identify genes essential for *M. capsulatus* growth. We observed a minor growth defect of CRISPRi induced cultures compared to uninduced cultures, but logarithmic growth kinetics and entry into stationary phase were similar between the conditions (Fig. 4B). The sgRNA library compositions in CRISPRi induced and noninduced samples 72 hours after selection were quantified by sgRNA amplicon sequencing and mapping to both the synthesized sgRNA library and the *M. capsulatus* genome (tables S6

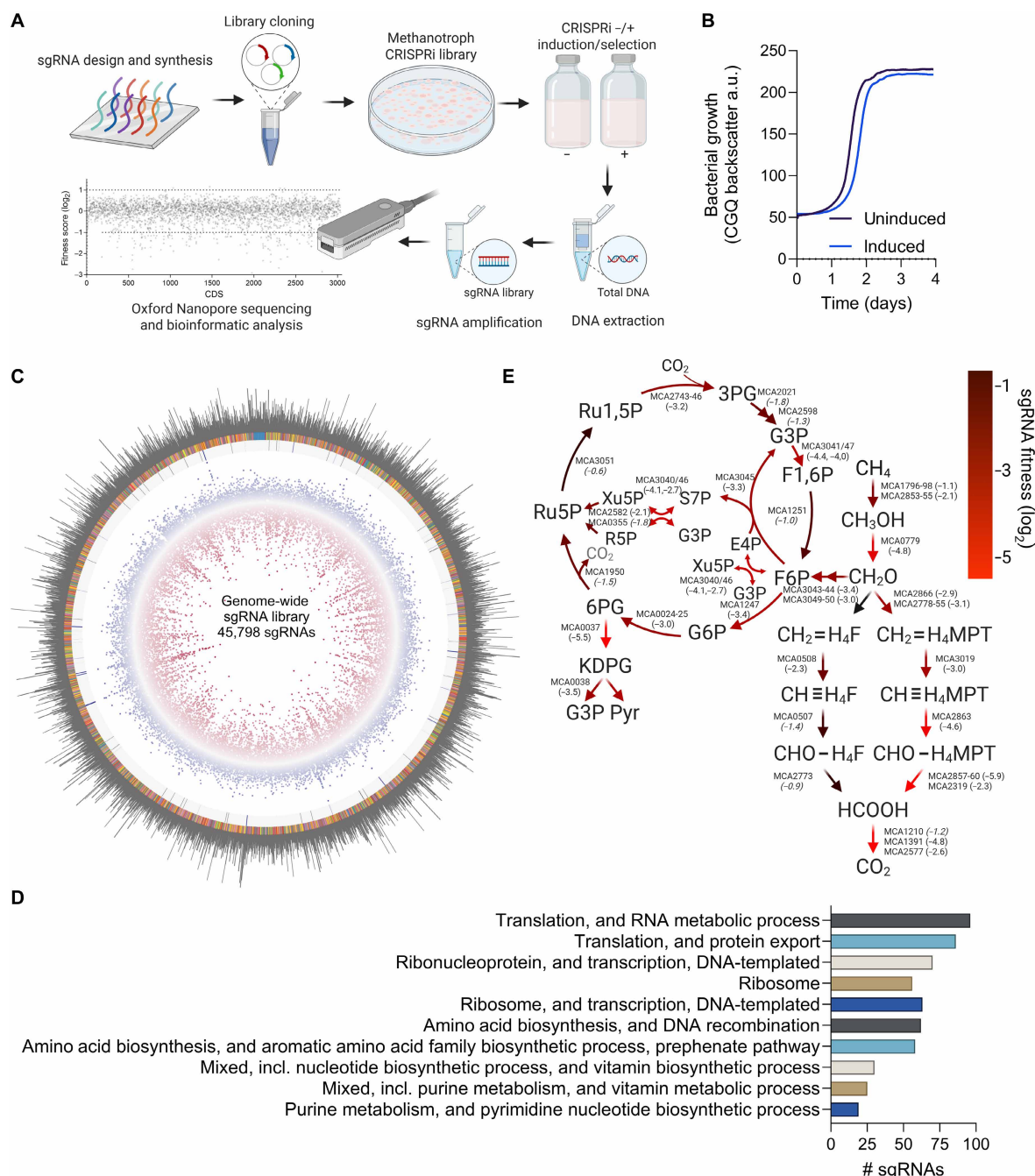


Fig. 4. A CRISPRi genome-wide screen identifies genes essential for CH₄-dependent growth. (A) Schematic workflow for methanotroph CRISPRi library construction. (B) Growth of the CRISPRi library cultivated in 20% CH₄ in air atmosphere with or without sgRNA and dCas9 induction. a.u., arbitrary units. (C) Circos plot of the methanotroph sgRNA library raw reads (outer ring) mapped to the 45,798 synthesized sgRNA oligos (rainbow ring with nontargeting sgRNAs in blue). Differential sgRNA fitness scores determined via DESeq2 between CRISPRi induced (n = 4) and uninduced (n = 4) samples during selection/cultivation with CH₄ as the sole carbon and energy source (fourth ring, blue dots: log₂ > 1; red dots: log₂ < 1). sgRNAs depleted in the screen correspond to genes with transcription repression causing a growth defect where those enriched correspond to genes with transcription repression causing a growth advantage. sgRNAs with statistically significant (P ≤ 0.01) differential fitness scores are indicated (blue squares, third ring). (D) ShinyGO enrichment analysis of the 699 genes with sgRNA log₂ fitness scores ≤ -1 and P ≤ 0.001. The 10 pathways with the lowest false discovery rates, sorted by the number of genes are shown. (E) CH₄ assimilatory and dissimilatory metabolic pathways predicted by sgRNA fitness scores. The mean sgRNA fitness scores are indicated in parentheses next to the gene loci that encode metabolic enzymes. Scores with P ≥ 0.01 are italicized. 3PG, 3-phosphoglycerate; G3P, glyceraldehyde 3-phosphate; F1,6P, fructose 1,6-bisphosphate; F6P, fructose 6-phosphate; E4P, erythrose 4-phosphate; Xu5P, xylulose 5-phosphate; S7P, sedoheptulose 7-phosphate; R5P, ribose 5-phosphate; Ru5P, ribulose 5-phosphate; R1,5P, ribulose 1,5-bisphosphate; G6P, glucose 6-phosphate; 6PG, 6-phosphogluconate; KDPG, 2-keto-3-deoxy-6-phosphogluconate; Pyr, pyruvate; CH₂=H₄F, methylene-tetrahydrofolate; CH₂=H₄MPT, methylene-tetrahydromethanopterin; CH=H₄F, methenyl-tetrahydrofolate; CH=H₄MPT, methenyl-tetrahydromethanopterin; CHO-H₄F, formyl-tetrahydrofolate; CHO-H₄MPT, formyl-tetrahydromethanopterin. Data in (B) represent the mean of two independent high-resolution cultivation experiments (n = 6). (A) and (E) were created in BioRender. Henard, C. (2026) <https://BioRender.com/57nyxu5>; <https://BioRender.com/p4flgui>.

to S13, read counts). We identified between 61.1 and 91.8% of the synthesized sgRNAs in the sample sequences, with increased coverage strongly correlated to sequencing depth (fig. S1, $R^2 = 0.93$). Although complete library coverage was not obtained in the screening samples with the sequencing depth achieved here, all gene targets were represented in the sequence data with 3027/3034 (99.8%) coding DNA sequences with ≥ 10 guides represented (tables S6 to S13). Comparing the sgRNA library composition between CRISPRi induced versus noninduced samples showed that 1019 sgRNAs (699 unique gene targets) were depleted ($\log_2$ fitness score ≤ 1 , $P \leq 0.001$) and 60 sgRNAs (47 unique gene targets) were enriched ($\log_2$ fitness score ≤ 1 , $P \leq 0.001$) after 72 hours of selection, presumably corresponding to a fitness defect or advantage phenotype associated with transcription repression of the sgRNA target gene (Fig. 4C and table S14). Enrichment analysis of the 699 unique genes with sgRNA $\log_2$ fitness scores ≤ -1 and probability values ≤ 0.001 indicated that many of these genes encode proteins associated with essential cellular processes like translation, transcription, ribosome biosynthesis, and amino acid and nucleotide biosynthesis (Fig. 4D and table S15). Enrichment analysis of the genes whose transcription repression resulted in a fitness advantage did not enrich any pathways using the available ShinyGO databases. Notably, 99.5% (398/400) of the non-targeting control sgRNAs did not have significant differential fitness scores ($\log_2 \leq -1$ or $\log_2 \geq 1$, $P \leq 0.001$) between the two conditions (table S14).

Many sgRNAs targeting the same gene or genes within operons exhibited similar fitness scores, highlighting the potential of the CRISPRi library and screening pipeline to identify functional gene targets (table S14). For example, sgRNAs targeting ribosomal protein operon genes MCA2368, MCA2371, MCA2372, and MCA2374 were consistently depleted under CRISPRi induction conditions with $\log_2$ fitness scores of -5.47 , -5.57 , -5.70 , and -6.37 , respectively, whereas sgRNAs targeting the cobalt transporter and cobalamin biosynthetic operon genes MCA2300, MCA2301, and MCA2302 were significantly enriched with $\log_2$ fitness scores of 3.10 , 2.70 , and 3.65 , respectively. However, significant fitness score variances between sgRNAs targeting the same gene were observed, suggesting that CRISPRi effectiveness in repressing transcription can vary depending on the sgRNA target sequence. For instance, the most depleted sgRNA under inducing conditions targets the MCA2374 gene encoding the translation elongation factor Tu ($\log_2$ fitness score of -6.37 ; table S14). However, the 14 other sgRNAs targeting MCA2374 showed variable fitness scores, none of which reached $P \leq 0.001$ statistical significance (table S14). This trend in variable sgRNA fitness scores is observed for most gene targets with at least one corresponding differential fitness score sgRNA, which is indicated by the high ratio of unique target genes within the total sgRNAs depleted ($699/1019 = 0.68$) or enriched ($47/60 = 0.78$). These data highlight the importance of including several sgRNAs for each gene in the CRISPRi library design and construction to increase the probability of identifying functional sgRNAs/genomes under selective conditions.

Several sgRNAs targeting known central CH_4 metabolic enzyme genes had significant negative fitness scores (Fig. 4E and table S14), including those targeting the particulate methane monooxygenase operons (*pmoCAB1*, MCA1796-98 and *pmoCAB2*, MCA2853-55), the calcium- and PQQ-dependent methanol dehydrogenase (MCA0779), and the RuMP cycle operons (MCA3040-45 and MCA3046-50). As expected, on the basis of our initial CRISPRi experiments (Fig. 3), sgRNAs targeting the *cbbLSQ* operon (MCA2743-46) were among

those with the lowest fitness scores. Furthermore, sgRNAs targeting the Entner-Doudoroff (ED) pathway *edd* and *eda* genes (MCA0037-38) were significantly depleted in the screen, consistent with biochemical data and genome-scale models predicting this pathway to be essential and the primary route for pyruvate biosynthesis in *M. capsulatus* Bath (5, 54, 55). Also, the depletion of sgRNAs targeting the tetrahydromethanopterin formaldehyde dissimilatory pathway corroborate flux balance model predictions that this is the primary route used by *M. capsulatus* for formaldehyde conversion to formate (54, 55). Similar to the MCA2375 (Tu)-targeting sgRNAs mentioned previously, fitness scores of sgRNAs targeting known or expected essential methanotroph genes varied. The individual *pmoCAB*- and *cbbLSQ*-targeting sgRNAs showed highly variable fitness scores, especially those targeting *pmoC*, *pmoA*, and the *cbbL* genes (fig. S2A). To confirm this variability, we tested the ability of three *pmoC*-targeting sgRNAs with variable fitness scores (sgRNA^{*pmoC1*}: $\log_2 = 0.34$; sgRNA^{*pmoC2*}: $\log_2 = -1.11$; sgRNA^{*pmoC3*}: $\log_2 = -2.10$) to inhibit *M. capsulatus* growth. The extent of bacterial growth inhibition was correlated to the respective sgRNA fitness scores (fig. S2B). The high-GC content of the *M. capsulatus* genome increases the possibility of sgRNA secondary structure formation, which may limit sgRNA targeting efficiency or render some nonfunctional (56, 57).

We next mapped the sgRNA target sequences from CRISPRi induced and noninduced cultures to the *M. capsulatus* Bath genome to determine overall sgRNA/gene fitness. Given the size of the sgRNA library and the observed variable fitness scores between the multiple sgRNAs per open reading frame, only those genes with overall positive or negative sgRNA fitness scores were expected to be identified in the analysis. We identified 233 genes whose transcription repression resulted in a fitness defect (sgRNAs depleted in the induced samples; $\log_2 \leq -1$, $P \leq 0.001$) and repression of 13 genes associated with a fitness advantage (sgRNAs enriched in the induced samples; $\log_2 \geq 1$, $P \leq 0.001$) (fig. S3A and table S16). Enrichment analysis of the 233 genes with significant sgRNA $\log_2$ fitness scores ≤ -1 linked the genes to the essential processes of translation, transcription, ribosome biosynthesis, and amino acid and nucleotide biosynthesis (fig. S3B and table S17), consistent with the analysis using the sgRNA reference library. The *edd* gene encoding the ED pathway 6-phosphogluconate dehydratase had the most negative differential fitness score ($\log_2 = -3.9$, $P = 1.63 \times 10^{-71}$; Table 1). Several putative essential genes are involved in antioxidant defenses, Fe-S cluster assembly/repair, and disulfide reduction, including superoxide dismutase (MCA2458; $\log_2 = -3.2$), a putative Rnf electron transport complex (MCA2893-2899; $\log_2 = -3.1$), ferredoxin (MCA2470; $\log_2 = -3.0$), and thioredoxin disulfide reductase (MCA1694; $\log_2 = -2.6$). These genes may be involved in balancing redox homeostasis during CH_4 oxidation (58, 59) and represent gene targets for downstream gene-function analysis. sgRNAs targeting genes in the cobalamin biosynthetic operon (MCA2290-MCA2302) had consistent positive fitness scores, and MCA2302, encoding a putative CbtB subunit of a cobalt transporter, was identified as significantly enriched ($\log_2 = 1.26$, $P = 3.9 \times 10^{-11}$) when mapping all sgRNAs to the *M. capsulatus* genome (Table 1 and table S16). As expected, on the basis of the variable fitness scores for sgRNAs targeting the pMMO and RuBisCO operons (fig. S2), only the *cbbS* (encoding the RuBisCO small subunit) and *cbbQ* (encoding a putative RuBisCO activase) genes with consistent sgRNA scores were associated with negative fitness scores when mapping to the genome. Another putative RuBisCO activase gene (MCA2754; $\log_2 = -3.1$, $P = 2.4 \times 10^{-22}$) may be essential for *M. capsulatus* growth on CH_4 . Evaluating the RuBisCO activase genes

Table 1. *M. capsulatus* genes with the greatest differential fitness scores identified via DESeq2 after sgRNA target sequence mapping to the genome.

Annotation	Locus tag	Differential fitness score (log ₂)	Differential fitness score <i>P</i> value
<i>edd</i> CDS	MCA0037	−3.89	1.6×10^{-71}
<i>ispC</i> CDS	MCA0573	−3.59	7.2×10^{-19}
<i>sodB</i> CDS	MCA2458	−3.22	2.3×10^{-18}
<i>rsxD</i> CDS	MCA2896	−3.21	3.4×10^{-23}
<i>dnaA</i> CDS	MCA3033	−3.14	7.0×10^{-22}
Nitric oxide reductase activation protein NorD CDS	MCA2754	−3.10	2.4×10^{-22}
Formylmethanofuran dehydrogenase subunit B CDS	MCA2860	−3.08	7.9×10^{-24}
<i>rsxG</i> CDS	MCA2897	−3.06	1.1×10^{-49}
Ferredoxin family protein CDS	MCA2470	−3.02	2.7×10^{-17}
<i>frr</i> CDS	MCA0570	−3.01	7.9×10^{-22}
<i>rpsB</i> CDS	MCA0567	−2.88	4.8×10^{-22}
HAD-IB family hydrolase CDS	MCA2087	−2.87	1.7×10^{-15}
<i>rsxA</i> CDS	MCA2893	−2.84	1.7×10^{-19}
Formylmethanofuran dehydrogenase subunit A CDS	MCA2859	−2.70	6.2×10^{-14}
<i>rpsS</i> CDS	MCA2368	−2.68	1.2×10^{-42}
<i>cysG</i> CDS	MCA2089	−2.63	2.3×10^{-11}
<i>tsf</i> CDS	MCA0568	−2.57	3.9×10^{-14}
<i>trxB</i> CDS	MCA1694	−2.56	2.7×10^{-23}
<i>pyrH</i> CDS	MCA0569	−2.53	1.8×10^{-13}
<i>asnB</i> CDS	MCA2127	−2.43	9.4×10^{-13}
<i>infB</i> CDS	MCA1315	−2.36	1.2×10^{-31}
<i>sufD</i> CDS	MCA0990	−2.35	5.5×10^{-22}
<i>rplD</i> CDS	MCA2371	−2.34	9.5×10^{-21}
<i>rplC</i> CDS	MCA2372	−2.28	2.3×10^{-23}
CbbQ/NirQ/NorQ/GpvN family protein CDS	MCA2746	−2.24	1.3×10^{-16}
OadG family protein CDS	MCA2481	−2.23	2.8×10^{-18}
<i>glyQ</i> CDS	MCA2049	−2.22	3.1×10^{-27}
<i>ispG</i> CDS	MCA2483	−2.19	9.0×10^{-17}
DNA binding protein CDS	MCA2958	−2.18	1.6×10^{-8}
ABC transporter substrate-binding protein CDS	MCA1278	−2.15	9.8×10^{-17}
Bifunctional cobalt-precorrin-7 (C(5))-methyltransferase/ cobalt-precorrin-6B (C(15))-methyltransferase CDS	MCA2296	−2.14	1.0×10^{-12}
Phosphoribosylaminoimidazolesuccinocarboxamide synthase CDS	MCA0021	−2.08	6.3×10^{-17}
Calcium/proton transporter CDS	MCA0545	−2.05	1.3×10^{-14}
OmpA family protein CDS	MCA1819	1.00	1.5×10^{-8}
YihY family inner membrane protein CDS	MCA0711	1.05	1.8×10^{-5}
<i>atpA</i> CDS	MCA0010	1.08	8.1×10^{-10}
<i>casSe</i> CDS	MCA0932	1.10	1.2×10^{-7}
Glutathione S-transferase N-terminal domain-containing protein CDS	MCA1959	1.11	2.9×10^{-7}
Hypothetical protein CDS	MCA1324	1.14	6.5×10^{-7}
Phosphoribulokinase CDS	MCA3051	1.15	7.6×10^{-11}
NUDIX hydrolase CDS	MCA0333	1.15	7.6×10^{-13}
PhoH family protein CDS	MCA1458	1.20	6.2×10^{-7}
CbtB-domain containing protein CDS	MCA2302	1.26	6.1×10^{-11}
MucB/RseB C-terminal domain-containing protein CDS	MCA1469	1.41	1.2×10^{-8}
CbtA family protein CDS	MCA2301	1.61	1.3×10^{-11}
RseA family anti-sigma factor CDS	MCA1470	1.62	1.1×10^{-6}

under differential growth conditions may provide insight into their roles in methanotrophic growth of *M. capsulatus*.

To validate our screening results, we individually targeted five genes with negative fitness scores involved in *M. capsulatus* central metabolism (*edd*, MCA0037, encoding phosphogluconate dehydratase of the ED pathway), DNA replication (*dnaA*, MCA3033, encoding the DNA replication initiator), redox metabolism (*fdx*, MCA2470, encoding a ferredoxin), ribosome biogenesis (*rpsB*, MCA0567, encoding the S2 protein component of the ribosome 30S subunit), and amino acid biosynthesis (*asnB*, MCA2127, encoding the asparagine synthase) and one gene with a positive fitness score (*cbtB*, MCA2302, encoding a putative cobalt transporter) (Table 1). Consistent with the screening results, high-resolution growth analysis of *M. capsulatus* strains expressing individual sgRNAs displayed significantly inhibited (*edd*, *dnaA*, *fdx*, *rpsB*, and *asnB*) or improved (*cbtB*) growth compared to uninduced controls or a nontargeting sgRNA (Figs. 5, A and F, and 6A). Like that observed when targeting the *cbbL* gene (Fig. 3), growth-inhibited strains began to expand after several days of selective pressure. We expect that suppressor mutations that bypass the CRISPRi machinery or gene expression defect may accumulate under selective pressure and dominate the population after extended cultivation, but additional inquiry is needed to identify the causal mutations. Collectively, these cultivation data validate that the CRISPRi system is effective for functional genomic screening and identification of genes required for CH₄-dependent growth.

Cultivation medium optimization improves methanotroph CH₄-dependent growth

Cobalt is an essential trace element; however, excess can limit bacterial growth by competing with iron and through the generation of reactive oxygen species (60, 61). On the basis of the growth improvement observed when transcription of the *CbtB* cobalt transporter was inhibited (Fig. 6A), we hypothesized that the trace amount of cobalt present in our NMS growth medium (1 μ M CoCl₂) was cytotoxic. Consistent with this, *M. capsulatus* cultivated in NMS with decreasing Co²⁺ concentrations displayed a dose-dependent growth improvement, primarily linked to reduced lag phase (Fig. 6B). *M. capsulatus* exhibited a 1.4-fold improvement in biomass yield within 48 hours in NMS medium with 0.01 μ M CoCl₂ compared to standard NMS [Fig. 6C, 0.65 versus 0.48 g dry cell weight (DCW)/g CH₄, $P = 0.006$], indicating

higher CH₄ flux to biomass under these conditions. The improved methanotroph growth kinetics in the optimized growth medium equates to a 4.7-fold improvement in methanotroph biomass productivity (Fig. 6D). Methanotroph biomass (i.e., single-cell protein) is currently generated industrially as a valuable CH₄-derived product used in the production of animal feed and agricultural fertilizer. Techno-economic analyses have identified methanotroph productivity as a primary cost driver of methane bioconversion processes (62, 63); thus, the optimized medium formulation identified here would be expected to significantly decrease single-cell protein production costs.

DISCUSSION

Our results demonstrate that the CRISPRi system developed here can be used for targeted transcription repression in phylogenetically diverse methanotrophic bacteria. Selective screening of a genome-wide sgRNA library during CH₄-dependent cultivation identified predicted and previously unidentified essential genes required for methanotrophic growth. The screen provides genetic evidence that the ED pathway is essential for CH₄ assimilation and the tetrahydromethanopterin pathway is the primary route for CH₄ dissimilation in *M. capsulatus*. In addition, the decreased fitness of sgRNAs targeting the RuBisCO and RuBisCO activase genes reiterates the essentiality of CO₂ assimilation for *M. capsulatus* growth on CH₄. Notably, our investigations led to the identification of an optimized growth medium for improved methanotroph biomass productivity.

We identified 233 genes (7.7% of open reading frames) as essential gene candidates in *M. capsulatus* Bath using CRISPRi screening when mapping sgRNA target sequences to the genome, a value on par with the percentage of essential and conditional essential genes in the *Escherichia coli* genome (9.8% of open reading frames) (64, 65) and many other phylogenetically diverse bacteria (66). When mapping to the sgRNA reference library, we identified 699 sgRNAs corresponding to unique gene targets with significantly differential fitness scores ($\log_2 \leq 1$, $P \leq 0.001$) between CRISPRi induced and uninduced conditions. The discrepancy between the total essential gene candidates identified between mapping to the genome or sgRNA reference library is likely linked to the high variance in fitness scores of sgRNAs targeting the same gene. Thus, it is crucial to include multiple sgRNAs for target genes in the initial CRISPRi library

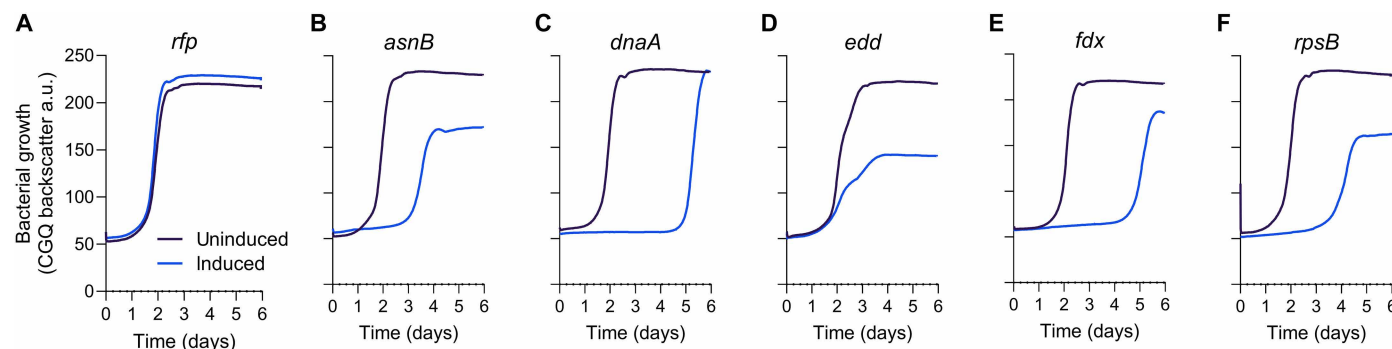


Fig. 5. CRISPRi targeted transcription repression of putative essential genes inhibits CH₄-dependent growth. (A to F) High-resolution growth curves of strains with sgRNAs targeting genes identified in the genome-wide screen as essential for CH₄-dependent growth with (blue line) or without (purple line) CRISPRi induction in NMS medium in serum vials with 20% CH₄ in air atmosphere. A strain expressing an sgRNA targeting mRFP was included as a control (A). Genes represent those encoding enzymes involved in amino acid biosynthesis [(B); *asnB*, MCA2127], DNA replication [(C); *dnaA*, MCA3033], CH₄ assimilation [(D); *edd*, MCA0037], redox homeostasis [(E); *fdx*, MCA2470], and ribosome biosynthesis [(F); *rpsB*, MCA0567]. a.u., arbitrary units. Data represent the mean of two independent high-resolution cultivation experiments ($n = 6$).

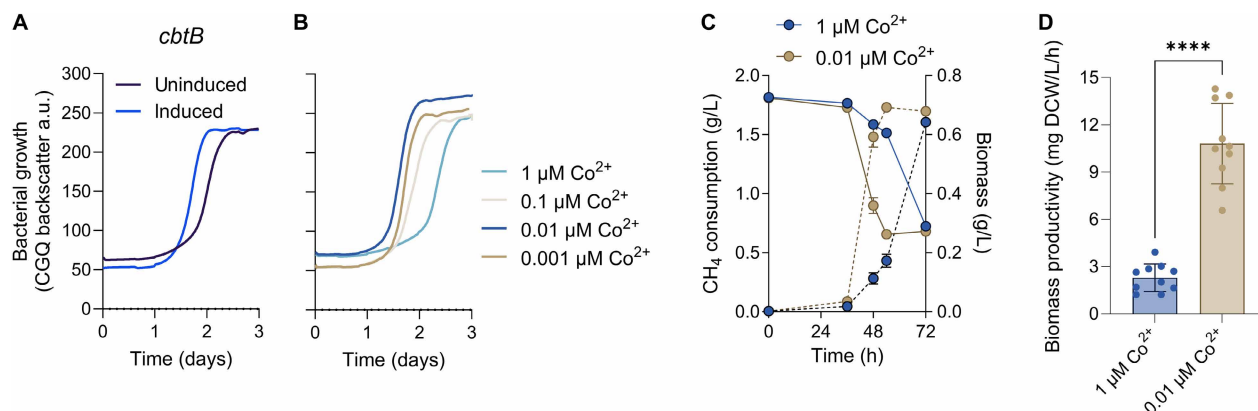


Fig. 6. CRISPRi screen-guided growth medium optimization improves methanotroph biomass productivity from CH₄. High-resolution growth curves of (A) a strain with an sgRNA targeting the *cbtB* gene with (blue line) or without (purple line) CRISPRi induction or (B) wild-type *M. capsulatus* in NMS medium with decreasing CoCl₂ in the trace elements. a.u., arbitrary units. (C) CH₄ consumption (solid lines) measured by gas chromatography and biomass accumulation (dashed lines) between wild-type cells cultured in high (1 μM CoCl₂; blue symbols) or low (0.01 μM CoCl₂; brown symbols) NMS medium in serum vials with 20% CH₄ in air atmosphere. (D) *M. capsulatus* biomass productivity in high or low CoCl₂ NMS medium at 48 hours. L, liter; h, hours. Data in (A) and (B) represent the mean of two independent high-resolution cultivation experiments ($n = 6$). Data in (C) and (D) represent the mean $\pm$ SEM from two independent experiments ($n = 6$ to 10). **** $P \leq 0.0001$.

design to increase the probability of identifying functional genes. However, multiple sgRNAs per gene can result in false negatives when mapping to the genome if there are variable target gene sgRNA fitness scores because the change in abundance of one or few sgRNAs is diluted by the other sgRNAs lacking differential abundance. Therefore, it is important to perform mapping to both the sgRNA reference library and the target genome. The sgRNA fitness scores from our screening results (table S14) may guide construction of much smaller genome-wide or targeted libraries consisting of the most effective sgRNAs for improved resolution in identifying functional genes.

Our results strongly support that the CRISPRi system is functional and can be used for forward genetic screening in methanotrophs, but there are limitations to this technology that likely resulted in false negatives in the selective screening process established herein. CRISPRi-dependent transcription repression is dynamic and rarely 100% effective. As such, decreased transcription may not be sufficient for selection when decreases in protein do not result in cell growth retardation or enhancement. This may be especially true for genes encoding biosynthetic enzymes where sufficient metabolic intermediate substrates could be made available via excretion or lysis from neighboring cells in the pooled library population (67, 68). Metabolite cross-feeding could explain why all genes expected to be essential under the methanotroph prototrophic growth conditions used here (e.g., amino acid, lipid, and vitamin biosynthetic genes) were not identified in the CRISPRi screen.

Collectively, the CRISPRi system and associated datasets herein will serve as valuable resources for hypothesis-driven inquiry into genes required for methanotrophy or alternative gene-function analyses. Ultimately, these CRISPRi tools can be coupled with other synthetic biology approaches to optimize methanotrophic strains for carbon- and energy-efficient CH₄ capture and conversion.

MATERIALS AND METHODS

Bacterial strains and cultivation

Bacterial strains used in this study are shown in table S18. *E. coli* strains were cultured in lysogeny broth (Lennox) with kanamycin

(25 μg/ml), spectinomycin (50 μg/ml), or gentamicin (10 μg/ml) for transformant selection. *M. capsulatus* Bath and *M. trichosporium* OB3b frozen stocks were streaked onto NMS solid medium and maintained in stainless steel gas chambers supplied with 20% CH₄ in the gas phase at 37° or 30°C, respectively, as previously described (45). Methanotroph strains were passaged weekly for 4 weeks (maximum passage = 5) after which time a new culture from frozen stock was initiated. Plasmids were transferred to methanotrophs via biparental mating by spreading equivalent biomass of S17-1λ *E. coli* and recipient methanotroph biomass on NMS mating agar mating plates and incubating in a 20% CH₄ atmosphere for 24 hours as previously described (45). After mating, methanotroph transformants harboring plasmids were selected on NMS medium containing spectinomycin (50 μg/ml; pJHCi and derivatives), gentamicin (10 μg/ml; pDS1), and/or kanamycin (50 μg/ml; pJHint). Methanotrophs were cultured in 150-ml vials containing 10 ml of NMS medium at 37°C (*M. capsulatus*) or 30°C (*M. trichosporium*) at 200-rpm orbital shaking. After inoculation with plate-derived biomass to an optical density at 600 nm (OD₆₀₀) of 0.01, vials were crimped with gray butyl stoppers to create gastight seals followed by CH₄ addition to the headspace via syringe to reach a final CH₄ concentration of 20% in air (v/v). Where applicable, sgRNA and/or dCas9 expression were induced with aTc (0.2 μg/ml). High-resolution bacterial growth was measured every 20 s by Cell Growth Quantifier (CGQ) optical sensors (Scientific Bioprocessing), and the data are presented as the mean arbitrary light backscatter units as directly detected by the sensors.

CRISPRi plasmid construction and sgRNA exchange

Primers and other synthetic DNA used in this study are shown in table S19. To construct a broad-host-range, single-plasmid CRISPRi system, we amplified a P_{mxsA}-sgRNA with an mRFP spacer sequence from the pgRNA plasmid (45) using primers oCAH267/289 and ligated it with pCAH01^{SPR}::Cas9 (45) linearized with primers oCAH230/231 via isothermal assembly using the HiFi Gibson Assembly Master Mix (New England Biolabs). We then amplified the dCas9 gene from plasmid pdCas9-bacteria (41) with oCAH952/1507 and assembled it with the above plasmid backbone amplified with

oCAH4/3, exchanging the Cas9 gene with dCas9 to generate plasmid pJHCi. The aTc-inducible suicide plasmid pJHint was constructed via isothermal assembly of the following six DNA fragments: (i) pUC origin of replication from pCAH01 amplified with oCAH403/404, (ii) 1 kilobase (kb) upstream homology arm amplified with oCAH405/406 from *M. capsulatus* genomic DNA, (iii) P_{tet} promoter/operator amplified from pCAH01 with oCAH407/408, (iv) *tetR*-kn^R cassette from pCAH01 amplified with oCAH409/410, (v) 1 kb downstream homology arm amplified with oCAH411/412 from *M. capsulatus* genomic DNA, and (vi) *oriT* origin of transfer amplified from pCAH01 with oCAH413/414. pJHint was then amplified with primers oCAH3/4 and assembled with the dCas9 gene fragment to generate plasmid pJHintdCas9. pJHintdCas9 was transferred to *M. capsulatus* Bath via conjugation, and transformants with the P_{tet} -dCas9 module integrated into an intergenic site between MCA_{tRNA-Ile} and MCA_{tRNA-Pro} were selected on NMS solid medium containing kanamycin (25 µg/ml). Successful integration was confirmed via polymerase chain reaction (PCR) using primers oCAH447/448 that flank the integration site. For sgRNA expression, we initially amplified the mRFP targeting sgRNA from pJHCi with primers oCAH1083/1081 and cloned it into pCAH01 amplified without a ribosomal binding site with primers oCAH4/119 to generate p01sgRNA^{mrfp}. Other aTc-inducible sgRNA expression plasmids were generated by exchanging the spacer sequence via isothermal assembly of a single-stranded oligo containing a unique spacer and the pCAH01sgRNA^{mrfp} amplified with primers oCAH1190/119.

Fluorescence measurements

pJHCi was transferred to *M. capsulatus* Bath or *M. trichosporium* OB3b harboring the pDS1 mRFP reporter plasmid via conjugation. Alternatively, pDS1 and p01sgRNA^{mrfp} were iteratively transferred to the *M. capsulatus* Bath strain with chromosomally integrated P_{tet} -dCas9. Overnight seed cultures of these strains were subcultured to OD₆₀₀ = 0.01 in liquid medium containing appropriate antibiotics with or without aTc induction and cultivated for 72 hours to an OD₆₀₀ of ~2.0. A 200-µl culture was transferred to a 96-well microplate, and mRFP1 fluorescence (ex_{532nm}, em_{588nm}, and gain = 80) and optical density (A_{600nm}) were measured with a BioTek Synergy Mx microplate reader. The data were normalized by dividing raw fluorescence reads by the optical density. mRFP expression was also monitored via fluorescence microscopy by transferring 5 µl of *M. capsulatus* logarithmic growth phase cultures with or without CRISPRi induction to a glass slide and mounted with a coverslip. RFP (ex: 575 to 640 nm; em: 629 nm) images were acquired using a Zeiss AXIO Imager M2 fluorescence microscope with 100X (oil) Plan-Apochromat (HC DIC) objective and processed with ZEN software.

Construction of the pLSQ and *cbbL* PAM mutant

The *M. capsulatus* Bath *cbbLSQ* operon (MCA2743-2746) including 150 bp upstream of *cbbL* presumed to contain the *cbbL* promoter was amplified from purified genomic DNA using primers oCAH1629 and oCAH1750. The amplicon was assembled with pDS1 amplified with primers oCAH1007 and oCAH16 to generate pLSQ. This plasmid was then linearized with primers oCAH1751 and oCAH16 and assembled with a synthetic fragment containing a *cbbL*:c.57C>T PAM mutation to generate pLSQ^{ΔPAM}. The pCAH01sgRNA^{cbbL} plasmid was transferred to *M. capsulatus* Bath harboring the pLSQ or pLSQ^{ΔPAM} strains, and growth was assessed with or without aTc induction using CGQ optical sensors.

Design, synthesis, and construction of the *M. capsulatus* genome-wide sgRNA library

Synbio Technologies designed and synthesized the sgRNA oligo pool following design rules to identify 15 PAM and target sequences per the predicted 3034 open reading frames in the NC_002977.6 RefSeq genome and to preferentially target the nontemplate DNA strand within the first 50% of the 5' end of the target gene. The sgRNA library consisted of 45,798 sgRNA oligos (120 bp) with each containing a unique spacer sequence and 5' and 3' homologous sequence for facile cloning into the p01sgRNA plasmid. The library also included 400 nontargeting sgRNAs with randomly generated spacers. The pooled sgRNA library was amplified via PCR using primers oCAH1416/1417 and Q5 polymerase (New England Biolabs) following the manufacturer's recommended reaction conditions for 20 cycles to limit biased amplification. The amplicons were assembled with p01sgRNA^{mrfp} amplified with oCAH1190/119 via isothermal assembly and transformed into high-efficiency DH10b competent cells (New England Biolabs). Before plating on selective medium, the competent cells were serially diluted to determine transformant colony-forming units (CFU/ml). A total of 1.02×10^7 DH10b transformants were obtained after transformation, three orders of magnitude greater than the synthesized library size. Twenty antibiotic-resistant transformants were screened via PCR and Sanger sequencing to determine the p01sgRNA^{mrfp} background, which was <5%. The transformants were then pooled, and the library miniprep using the Monarch Plasmid Miniprep Kit (New England Biolabs). The mixed p01sgRNA library was then transferred to chemically competent S17 *E. coli* obtaining 1.15×10^6 S17 transformants, which were conjugated with the *M. capsulatus* Bath strain with integrated P_{tet} -dCas9 to generate the methanotroph CRISPRi library. *M. capsulatus* Bath transformants were quantified via serial dilution plating and pooled for genomic DNA extraction and long-term -80°C storage. A total of 6.8×10^7 transconjugants were obtained, 1000X the amount of unique sgRNAs in the synthesized library. Library composition was determined via ONT amplicon sequencing as described below. The pooled methanotroph CRISPRi library was cultivated in liquid NMS, pelleted, resuspended in NMS containing 8% dimethyl sulfoxide (DMSO) to OD₆₀₀ = 5 ($\sim 5 \times 10^8$ CFU/ml), and aliquoted for cryostorage. To ensure that the library was not diluted by routine streaking and passaging on solid medium, individual 1-ml aliquots of the library was thawed on ice, added to 20 ml of NMS in a serum vial, and cultivated overnight at 37°C. The library seed culture was then diluted to OD₆₀₀ = 0.01 in fresh NMS with or without the aTc inducer for selective screening.

ONT sequencing and bioinformatic analysis

The sgRNA library was cultivated in liquid NMS with or without CRISPRi induction. After 72 hours of selection, genomic DNA was extracted from an OD = 2 equivalent cell pellet using the Zymo DNA extraction kit following the manufacturer's protocol. Fifty nanograms of the pooled sgRNA library was used as a template to amplify an ~250-bp amplicon containing the sgRNA sequence with Q5 polymerase using primers oCAH1416/1417 and the same reaction conditions as above (20 cycles). The amplicons were barcoded with the native barcoding kit 24 V14 following the manufacturer's protocol (ONT) and sequenced using a MinION sequencing device with high accuracy basecalling. Fastq sequences were then trimmed and filtered using Geneious Prime software. The 20-bp sgRNA target sequences were extracted from the trimmed and filtered sequence reads using a custom Python script to process ONT sequence data stored as compressed .fasta.gz files. The script performs a

two-step search for the constant sgRNA sequences (TATTTTACCACTCCCTATCAGTGAT, TTTTAGAGCTAGAAATAGCAAGTT or their reverse complements ATCACTGATAGGGAGTGGTAAAA-TA, AACTTGCTATTTCTAGCTCTAAAAC) flanking the target sequence and extracts the adjacent 5' or 3' 20-bp target sequence from the read dependent on the identified query sequence/read strand orientation. The script uses pairwise alignment to refine the search results, ensuring accurate extraction of relevant sequences. After extraction, the sgRNA target sequences were aligned to either a custom reference library containing the 45,798 sgRNA library sequences or the *M. capsulatus* Bath genome (NC_002977.6), mapped reads were quantified and normalized as sgRNA target sequence per million (TPM), and the fitness score (sgRNA target sequence differential abundance between induced and uninduced samples) was determined following a DESeq2 workflow using Geneious software. Enrichment analysis of the most differentially abundant sgRNA targets was performed using ShinyGO 0.82 using the STRING database (69).

Co²⁺-deplete cultivation, gas chromatography, and biomass productivity calculations

Wild-type *M. capsulatus* solid medium-derived biomass was inoculated into acid-washed serum vials containing 10 ml of NMS medium with decreasing CoCl₂ in the trace elements to OD₆₀₀ = 0.01, and growth was measured with CGQ optical sensors. All medium components were made with high-performance liquid chromatography (HPLC)-grade water in acid-washed glassware. The CH₄ in the headspace was determined 10 min after gas addition and every 24 hours during methanotroph cultivation using an SRI 8610c gas chromatograph with a TCD (thermal conductivity detector) and an FID (flame ionization detector) (SRI Instruments). CH₄ was quantified as % atmospheric composition by comparison to known standards and converted to weight using the known headspace volume and specific gravity (0.622 mg/ml at 37°C). DCW was determined by converting OD₆₀₀ to DCW using a formula empirically determined by comparing OD₆₀₀ measurements to colony-forming units during serum vial cultivation (OD₆₀₀ = 1 is equivalent to 0.254 ± 0.0249 g DCW/liter).

Illustrations

Data were graphed using GraphPad Prism 10 software. Circos plots were generated using Circa (<https://circa.omgenomics.com/>).

Statistical analysis

Statistical analysis was performed using GraphPad Prism 10 software. The data between two groups were analyzed using unpaired *t* tests. Data were considered statistically significant when *P* ≤ 0.05.

Supplementary Materials

The PDF file includes:

Figs. S1 to S3

Legends for tables S1 to S17

Tables S18 and S19

Other Supplementary Material for this manuscript includes the following:

Tables S1 to S17

REFERENCES

1. S. Guerrero-Cruz, A. Vaksmaa, M. A. Horn, H. Niemann, M. Pijuan, A. Ho, Methanotrophs: Discoveries, environmental relevance, and a perspective on current and future applications. *Front. Microbiol.* **12**, 678057 (2021).
2. M. E. Lidstrom, Direct methane removal from air by aerobic methanotrophs. *Cold Spring Harb. Perspect. Biol.* **16**, a041671 (2024).
3. F. J. Tucci, A. C. Rosenzweig, Direct methane oxidation by copper- and iron-dependent methane monooxygenases. *Chem. Rev.* **124**, 1288–1320 (2024).
4. M. G. Kalyuzhnaya, O. A. Gomez, J. C. Murrell, in *Taxonomy, Genomics and Ecophysiology of Hydrocarbon-Degrading Microbes*, T. J. McGenity, Ed. (Springer International Publishing, 2019), pp. 1–34.
5. T. Strom, T. Ferenci, J. R. Quayle, The carbon assimilation pathways of *Methylococcus capsulatus*, *Pseudomonas methanica* and *Methylosinus trichosporium* (OB3B) during growth on methane. *Biochem. J.* **144**, 465–476 (1974).
6. J. Heptinstall, J. R. Quayle, Pathways leading to and from serine during growth of *Pseudomonas* AM1 on C1 compounds or succinate. *Biochem. J.* **117**, 563–572 (1970).
7. A. J. Lawrence, M. B. Kemp, J. R. Quayle, Synthesis of cell constituents by methane-grown *Methylococcus capsulatus* and *Methanomonas methanooxidans*. *Biochem. J.* **116**, 631–639 (1970).
8. L. Chistoserdova, J. A. Vorholt, R. K. Thauer, M. E. Lidstrom, C1 transfer enzymes and coenzymes linking methylotrophic bacteria and methanogenic Archaea. *Science* **281**, 99–102 (1998).
9. J. A. Vorholt, Cofactor-dependent pathways of formaldehyde oxidation in methylotrophic bacteria. *Arch. Microbiol.* **178**, 239–249 (2002).
10. M. G. Kalyuzhnaya, S. Yang, O. N. Rozova, N. E. Smalley, J. Clubb, A. Lamb, G. A. N. Gowda, D. Raftery, Y. Fu, F. Bringel, S. Vuilleumier, D. A. C. Beck, Y. A. Trotsenko, V. N. Khmelenina, M. E. Lidstrom, Highly efficient methane biocatalysis revealed in a methanotrophic bacterium. *Nat. Commun.* **4**, 2785 (2013).
11. S. Vuilleumier, V. N. Khmelenina, F. Bringel, A. S. Reshetnikov, A. Lajus, S. Mangelot, Z. Rouy, H. J. M. Op den Camp, M. S. M. Jetten, A. A. Dispirito, P. Dunfield, M. G. Klotz, J. D. Semrau, L. Y. Stein, V. Barbe, C. Médigue, Y. A. Trotsenko, M. G. Kalyuzhnaya, Genome sequence of the haloalkaliphilic methanotrophic bacterium *Methylomicrobium alcaliphilum* 20Z. *J. Bacteriol.* **194**, 551–552 (2012).
12. A. Mateos-Rivera, T. Islam, I. P. G. Marshall, L. Schreiber, L. Øvreås, High-quality draft genome of the methanotroph *Methylovulum psychrotolerans* Str. HV10-M2 isolated from plant material at a high-altitude environment. *Stand. Genomic Sci.* **13**, 10 (2018).
13. K. K. Miroshnikov, A. Didriksen, D. G. Naumoff, M. Huntemann, A. Clum, M. Pillay, K. Palaniappan, N. Varghese, N. Mikhailova, S. Mukherjee, T. B. K. Reddy, C. Daum, M. Shapiro, N. Ivanova, N. Kyrpides, T. Woyke, S. N. Dedysh, M. M. Svenning, Draft genome sequence of *Methylocapsa palarum* NE2^T, an obligate methanotroph from subarctic soil. *Genome Announc.* **5**, e00504-17 (2017).
14. M. G. Kalyuzhnaya, A. E. Lamb, T. L. McTaggart, I. Y. Oshkin, N. Shapiro, T. Woyke, L. Chistoserdova, Draft genome sequences of gammaproteobacterial methanotrophs isolated from Lake Washington sediment. *Genome Announc.* **3**, e00103-15 (2015).
15. R. Hamilton, K. D. Kits, V. A. Ramonovskaya, O. N. Rozova, H. Yurimoto, H. Iguchi, V. N. Khmelenina, Y. Sakai, P. F. Dunfield, M. G. Klotz, C. Knief, H. J. M. Op den Camp, M. S. M. Jetten, F. Bringel, S. Vuilleumier, M. M. Svenning, N. Shapiro, T. Woyke, Y. A. Trotsenko, L. Y. Stein, M. G. Kalyuzhnaya, Draft genomes of gammaproteobacterial methanotrophs isolated from terrestrial ecosystems. *Genome Announc.* **3**, e00515-15 (2015).
16. F. D. Orata, K. D. Kits, L. Y. Stein, Complete genome sequence of *Methylomonas denitrificans* strain FJG1, an obligate aerobic methanotroph that can couple methane oxidation with denitrification. *Genome Announc.* **6**, e00276-18 (2018).
17. I. Y. Oshkin, R. Z. Suleimanov, V. N. Khmelenina, A. V. Mardanov, N. V. Pimenov, S. N. Dedysh, Complete genome sequence of *Methylococcus capsulatus* MIR, a methanotroph capable of growth on methanol. *Microbiol. Resour. Announc.* **11**, e0054222 (2022).
18. L. Y. Stein, S. Yoon, J. D. Semrau, A. A. Dispirito, A. Crombie, J. C. Murrell, S. Vuilleumier, M. G. Kalyuzhnaya, H. J. M. Op den Camp, F. Bringel, D. Bruce, J. F. Cheng, A. Copeland, L. Goodwin, S. Han, L. Hauser, M. S. M. Jetten, A. Lajus, M. L. Land, A. Lapidus, S. Lucas, C. Médigue, S. Pitluck, T. Woyke, A. Zeytun, M. G. Klotz, Genome sequence of the obligate methanotroph *Methylosinus trichosporium* strain OB3b. *J. Bacteriol.* **192**, 6497–6498 (2010).
19. S. J. Van Dien, Y. Okubo, M. T. Hough, N. Korotkova, T. Taitano, M. E. Lidstrom, Reconstruction of C₃ and C₄ metabolism in *Methylobacterium extorquens* AM1 using transposon mutagenesis. *Microbiology* **149**, 601–609 (2003).
20. C. J. Marx, B. N. O'Brien, J. Breeze, M. E. Lidstrom, Novel methylotrophy genes of *Methylobacterium extorquens* AM1 identified by using transposon mutagenesis including a putative dihydromethanopterin reductase. *J. Bacteriol.* **185**, 669–673 (2003).
21. K. E. Lee, S. Stone, P. M. Goodwin, B. W. Holloway, Characterization of transposon insertion mutants of *Methylobacterium extorquens* AM1 (*Methylobacterium* strain AM1) which are defective in methanol oxidation. *Microbiology* **137**, 895–904 (1991).
22. L. Chistoserdova, in *Microbial Growth on C1 Compounds*, M. E. Lidstrom, F. R. Tabita, Eds. (Springer Netherlands, 1996), pp. 16–24.
23. J. A. Vorholt, C. J. Marx, M. E. Lidstrom, R. K. Thauer, Novel formaldehyde-activating enzyme in *Methylobacterium extorquens* AM1 required for growth on methanol. *J. Bacteriol.* **182**, 6645–6650 (2000).

24. D. N. Nunn, M. E. Lidstrom, Phenotypic characterization of 10 methanol oxidation mutant classes in *Methylobacterium* sp. strain AM1. *J. Bacteriol.* **166**, 591–597 (1986).
25. A. L. Springer, H. H. Chou, W. H. Fan, E. Lee, M. E. Lidstrom, Methanol oxidation mutants in *Methylobacterium extorquens* AM1: Identification of new genetic complementation groups. *Microbiology* **141** (Pt. 11), 2985–2993 (1995).
26. S. Whitta, M. I. Sinclair, B. W. Holloway, Transposon mutagenesis in *Methylobacterium* AM1 (*Pseudomonas* AM1). *Microbiology* **131**, 1547–1549 (1985).
27. P. Roszczenko-Jasińska, H. N. Vu, G. A. Subuyuj, R. V. Crisostomo, J. Cai, N. F. Lien, E. J. Clippard, E. M. Ayala, R. T. Ngo, F. Yarza, J. P. Wingett, C. Raghuraman, C. A. Hoebler, N. C. Martinez-Gomez, E. Skovran, Gene products and processes contributing to lanthanide homeostasis and methanol metabolism in *Methylorubrum extorquens* AM1. *Sci. Rep.* **10**, 12663 (2020).
28. A. M. Ochser, M. Christen, L. Hemmerle, R. Peyraud, B. Christen, J. A. Vorholt, Transposon sequencing uncovers an essential regulatory function of phosphoribulokinase for methylotrophy. *Curr. Biol.* **27**, 2579–2588.e6 (2017).
29. J. D. Groom, S. M. Ford, M. W. Pesesky, M. E. Lidstrom, A mutagenic screen identifies a TonB-dependent receptor required for the lanthanide metal switch in the type I methanotroph "*Methylotuvimicrobium buryatense*" 5GB1C. *J. Bacteriol.* **201**, e00120-19 (2019).
30. E. Williams, M. A. Shimmin, B. W. Bainbridge, Mutation in the obligate methylotrophs *Methylococcus capsulatus* and *Methylomonas albus*. *FEMS Microbiol. Lett.* **2**, 293–296 (1977).
31. P. L. Sharpe, D. Dicosimo, M. D. Bosak, K. Knoke, L. Tao, Q. Cheng, R. W. Ye, Use of transposon promoter-probe vectors in the metabolic engineering of the obligate methanotroph *Methylomonas* sp. strain 16a for enhanced C40 carotenoid synthesis. *Appl. Environ. Microbiol.* **73**, 1721–1728 (2007).
32. M. Cheng, D. Pei, L. He, Q. Fei, X. Yan, Cre/lox-mediated CRISPRi library reveals core genome of a type I methanotroph *Methylotuvimicrobium buryatense* 5GB1C. *Appl. Environ. Microbiol.* **89**, e0188322 (2023).
33. J. S. Hawkins, S. Wong, J. M. Peters, R. Almeida, L. S. Qi, Targeted transcriptional repression in bacteria using CRISPR interference (CRISPRi). *Methods Mol. Biol.* **1311**, 349–362 (2015).
34. V. de Bakker, X. Liu, A. M. Bravo, J.-W. Veening, CRISPRi-seq for genome-wide fitness quantification in bacteria. *Nat. Protoc.* **17**, 252–281 (2022).
35. J. S. Hawkins, M. R. Silvis, B.-M. Koo, J. M. Peters, H. Osadnik, M. Jost, C. C. Hearne, J. S. Weissman, H. Todor, C. A. Gross, Mismatch-CRISPRi reveals the co-varying expression-fitness relationships of essential genes in *Escherichia coli* and *Bacillus subtilis*. *Cell Syst.* **11**, 523–535.e9 (2020).
36. L. Yao, K. Shabestary, S. M. Björk, J. Asplund-Samuelsson, H. N. Joensson, M. Jahn, E. P. Hudson, Pooled CRISPRi screening of the cyanobacterium *Synechocystis* sp. PCC 6803 for enhanced industrial phenotypes. *Nat. Commun.* **11**, 1666 (2020).
37. J. Shin, J. Bae, H. Lee, S. Kang, S. Jin, Y. Song, S. Cho, B.-K. Cho, Genome-wide CRISPRi screen identifies enhanced autolithotrophic phenotypes in acetogenic bacterium *Eubacterium limosum*. *Proc. Natl. Acad. Sci. U.S.A.* **120**, e2216244120 (2023).
38. X.-H. Mo, H. Zhang, T.-M. Wang, C. Zhang, C. Zhang, X.-H. Xing, S. Yang, Establishment of CRISPR interference in *Methylorubrum extorquens* and application of rapidly mining a new phytoene desaturase involved in carotenoid biosynthesis. *Appl. Microbiol. Biotechnol.* **104**, 4515–4532 (2020).
39. T. Wang, C. Guan, J. Guo, B. Liu, Y. Wu, Z. Xie, C. Zhang, X.-H. Xing, Pooled CRISPR interference screening enables genome-scale functional genomics study in bacteria with superior performance. *Nat. Commun.* **9**, 2475 (2018).
40. W. Jiang, P. Oikonomou, S. Tavao, Comprehensive genome-wide perturbations via CRISPR adaptation reveal complex genetics of antibiotic sensitivity. *Cell* **180**, 1002–1017.e31 (2020).
41. L. S. Qi, M. H. Larson, L. A. Gilbert, J. A. Doudna, J. S. Weissman, A. P. Arkin, W. A. Lim, Repurposing CRISPR as an RNA-guided platform for sequence-specific control of gene expression. *Cell* **152**, 1173–1183 (2013).
42. A. L. Enright, W. J. Heelan, R. D. Ward, J. M. Peters, CRISPRi functional genomics in bacteria and its application to medical and industrial research. *Microbiol. Mol. Biol. Rev.* **88**, e0017022 (2024).
43. B. L. Rumah, B. H. Claxton Stevens, J. E. Yeboah, C. E. Stead, E. L. Harding, N. P. Minton, Y. Zhang, In vivo genome editing in type I and II methanotrophs using a CRISPR/Cas9 system. *ACS Synth. Biol.* **12**, 544–554 (2023).
44. J. Jeong, T. H. Kim, N. Jang, M. Ko, S. K. Kim, J. I. Baek, G. Emelianov, E. Rha, K. K. Kwon, H. Kim, E. Y. Lee, D.-H. Lee, H. Lee, S.-G. Lee, A highly efficient and versatile genetic engineering toolkit for a methanotroph-based biorefinery. *Chem. Eng. J.* **453**, 139911 (2023).
45. T. Tapscott, M. T. Guarnieri, C. A. Henard, Development of a CRISPR/Cas9 system for *Methylococcus capsulatus* in vivo gene editing. *Appl. Environ. Microbiol.* **85**, e00340-19 (2019).
46. S. Nath, J. M. Henard, C. A. Henard, Optimized tools and methods for methanotroph genome editing. *Methods Mol. Biol.* **2489**, 421–434 (2022).
47. G. Emelianov, D.-U. Song, N. Jang, M. Ko, S. K. Kim, E. Rha, J. Shin, K. K. Kwon, H. Kim, D.-H. Lee, H. Lee, S.-G. Lee, Engineered *Methylococcus capsulatus* Bath for efficient methane conversion to isoprene. *Bioresour. Technol.* **393**, 130098 (2024).
48. E. H. Bhat, J. M. Henard, S. A. Lee, D. McHaffey, M. S. Ravulapati, E. V. Rogers, L. Yu, D. Skiles, C. A. Henard, Construction of a broad-host-range Anderson promoter series and particulate methane monooxygenase promoter variants expand the methanotroph genetic toolbox. *Synth. Syst. Biotechnol.* **9**, 250–258 (2024).
49. C. A. Henard, C. Wu, W. Xiong, J. M. Henard, B. Davidheiser-Kroll, F. D. Orata, M. T. Guarnieri, Ribulose-1,5-bisphosphate carboxylase/oxygenase (RubisCO) is essential for growth of the methanotroph *Methylococcus capsulatus* strain Bath. *Appl. Environ. Microbiol.* **87**, e0088121 (2021).
50. N. Politi, L. Pasotti, S. Zucca, M. Casanova, G. Micoli, M. G. Cusella De Angelis, P. Magni, Half-life measurements of chemical inducers for recombinant gene expression. *J. Biol. Eng.* **8**, 5 (2014).
51. H. Zhao, Y. Sun, J. M. Peters, C. A. Gross, E. C. Garner, J. D. Helmann, Depletion of undecaprenyl pyrophosphate phosphatases disrupts cell envelope biogenesis in *Bacillus subtilis*. *J. Bacteriol.* **198**, 2925–2935 (2016).
52. S. H. Sternberg, S. Redding, M. Jinek, E. C. Greene, J. A. Doudna, DNA interrogation by the CRISPR RNA-guided endonuclease Cas9. *Nature* **507**, 62–67 (2014).
53. X. Wu, D. A. Scott, A. J. Kriz, A. C. Chiu, P. D. Hsu, D. B. Dadon, A. W. Cheng, A. E. Trevino, S. Konermann, S. Chen, R. Jaenisch, F. Zhang, P. A. Sharp, Genome-wide binding of the CRISPR endonuclease Cas9 in mammalian cells. *Nat. Biotechnol.* **32**, 670–676 (2014).
54. A. Gupta, A. Ahmad, D. Chothwe, M. K. Madhu, S. Srivastava, V. K. Sharma, Genome-scale metabolic reconstruction and metabolic versatility of an obligate methanotroph *Methylococcus capsulatus* str. Bath. *PeerJ* **7**, e6685 (2019).
55. C. Lieven, L. A. H. Petersen, S. B. Jørgensen, K. V. Gernaey, M. J. Herrgard, N. Sonnenschein, A genome-scale metabolic model for *Methylococcus capsulatus* (Bath) suggests reduced efficiency electron transfer to the particulate methane monooxygenase. *Front. Microbiol.* **9**, 2947 (2018).
56. S. Riesenberger, N. Helmbrecht, P. Kanis, T. Maricic, S. Pääbo, Improved gRNA secondary structures allow editing of target sites resistant to CRISPR-Cas9 cleavage. *Nat. Commun.* **13**, 489 (2022).
57. E. A. Moreb, M. D. Lynch, A meta-analysis of gRNA library screens enables an improved understanding of the impact of gRNA folding and structural stability on CRISPR-Cas9 activity. *CRISPR J.* **5**, 146–154 (2022).
58. Y. Jouanneau, H. S. Jeong, N. Hugo, C. Meyer, J. C. Willison, Overexpression in *Escherichia coli* of the *rnf* genes from *Rhodobacter capsulatus*—Characterization of two membrane-bound iron-sulfur proteins. *Eur. J. Biochem.* **251**, 54–64 (1998).
59. A. B. Alleman, J. W. Peters, Mechanisms for generating low potential electrons across the metabolic diversity of nitrogen-fixing bacteria. *Appl. Environ. Microbiol.* **89**, e0037823 (2023).
60. M. P. Thorgersen, D. M. Downs, Cobalt targets multiple metabolic processes in *Salmonella enterica*. *J. Bacteriol.* **189**, 7774–7781 (2007).
61. S. Leonard, P. M. Gannett, Y. Rojanasakul, D. Schwegler-Berry, V. Castranova, V. Vallyathan, X. Shi, Cobalt-mediated generation of reactive oxygen species and its possible mechanism. *J. Inorg. Biochem.* **70**, 239–244 (1998).
62. Q. Fei, B. Liang, L. Tao, E. C. D. Tan, R. Gonzalez, C. Henard, M. Guarnieri, Biological valorization of natural gas for the production of lactic acid: Techno-economic analysis and life cycle assessment. *Biochem. Eng. J.* **158**, 107500 (2020).
63. S. H. El Abbadi, E. D. Sherwin, A. R. Brandt, S. P. Luby, C. S. Criddle, Displacing fishmeal with protein derived from stranded methane. *Nat. Sustain.* **5**, 47–56 (2022).
64. A. R. Joyce, J. L. Reed, A. White, R. Edwards, A. Osterman, T. Baba, H. Mori, S. A. Lesely, B. Ø. Palsøn, S. Agarwalla, Experimental and computational assessment of conditionally essential genes in *Escherichia coli*. *J. Bacteriol.* **188**, 8259–8271 (2006).
65. T. Baba, T. Ara, M. Hasegawa, Y. Takai, Y. Okumura, M. Baba, K. A. Datsenko, M. Tomita, B. L. Wanner, H. Mori, Construction of *Escherichia coli* K-12 in-frame, single-gene knockout mutants: The Keio collection. *Mol. Syst. Biol.* **2**, MSB4100050 (2006).
66. G. Fernández-García, P. Valdés-Chiara, P. Villazán-Gamonal, S. Alonso-Fernández, A. Manteca, Essential genes discovery in microorganisms by transposon-directed sequencing (Tn-Seq): Experimental approaches, major goals, and future perspectives. *Int. J. Mol. Sci.* **25**, 11298 (2024).
67. G. D'Souza, S. Shitut, D. Preussger, G. Yousif, S. Waschina, C. Kost, Ecology and evolution of metabolic cross-feeding interactions in bacteria. *Nat. Prod. Rep.* **35**, 455–488 (2018).
68. E. C. Seth, M. E. Taga, Nutrient cross-feeding in the microbial world. *Front. Microbiol.* **5**, 350 (2014).
69. S. X. Ge, D. Jung, R. Yao, ShinyGO: A graphical gene-set enrichment tool for animals and plants. *Bioinformatics* **36**, 2628–2629 (2020).

Acknowledgments: We would like to thank T. Cox in the Azad laboratory for computational support with the initial sgRNA library design. **Funding:** This work was supported by the BioDiscovery Institute Seed Grant (C.A.H. and R.K.A.), Department of Energy via contract

#W911SR-14-2-0001 with the MSI STEM Research & Development Consortium (C.A.H.), National Science Foundation MCB award #2225776 (C.A.H.), and National Science Foundation MCB CAREER award #2441696 (C.A.H.). **Author contributions:** Conceptualization: C.A.H. Methodology: J.M.H., Y.-C.Y., R.K.A., and C.A.H. Software: R.K.A., and C.A.H. Validation: J.M.H., Y.-C.Y., S.A.L., and C.A.H. Formal analysis: J.M.H., R.K.A., D.S., and C.A.H. Investigation: J.M.H., Y.-C.Y., R.K.A., S.A.L., and C.A.H. Resources: R.K.A. and C.A.H. Data curation: J.M.H., D.S., and C.A.H. Writing—original draft: D.S. and C.A.H. Writing—review and editing: J.M.H., Y.-C.Y., R.K.A., and C.A.H. Visualization: J.M.H. and C.A.H. Supervision: R.K.A. and C.A.H. Project administration: R.K.A. and C.A.H. Funding acquisition: R.K.A. and C.A.H. **Competing interests:** The authors declare that they have no competing interests. **Data, code, and materials availability:** All

data and code needed to evaluate and reproduce the results in the paper are present in the paper and/or the Supplementary Materials. The code used for sgRNA target sequence extraction can be downloaded from Zenodo: <https://doi.org/10.5281/zenodo.18227249>. The materials presented in this study can be provided by the Henard Laboratory upon reasonable request. Requests for materials should be submitted to: Calvin.Henard@unt.edu.

Submitted 28 October 2025

Accepted 19 March 2026

Published 24 April 2026

10.1126/sciadv.aed4234

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Sci. Adv. **12** (17), eaed4234. DOI: 10.1126/sciadv.aed4234

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